Exp.no : 07 **EDA - QUANTITATIVE AND QUALITATIVE ANALYSIS**

AIM:

To experiment and understand EDA-Quantitative and Qualitative analysis.

PYTHON CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set visualization style
sns.set_theme(style="whitegrid")
plt.rcParams["figure.figsize"] = (8, 5)

data = {
    "Student_ID": range(1, 16),

    "Name": [
        "Arun", "Bala", "Chitra", "Divya", "Eshan",
        "Farah", "Gokul", "Hari", "Isha", "Jaya",
        "Kiran", "Latha", "Manoj", "Nisha", "Omkar"
    ],

    "Age": [
        20, 21, 20, 22, 21,
        20, 23, 21, 22, 20,
        21, 22, 20, 21, 22
    ],

    "Gender": [
        "Male", "Male", "Female", "Female", "Male",
        "Female", "Male", "Male", "Female", "Female",
        "Male", "Female", "Male", "Female", "Male"
    ],

    "Department": [
        "CSE", "ECE", "CSE", "EEE", "CSE",
        "ECE", "EEE", "CSE", "ECE", "EEE",
        "CSE", "ECE", "CSE", "EEE", "ECE"
    ],
```

```
    "Study_Hours": [
        5, 3, 6, 2, 4,
        6, 1, 4, 7, 3,
        5, 6, 4, 5, 3
    ],

    "Attendance": [
        95, 80, 98, 75, 85,
        92, 60, 82, 96, 78,
        90, 94, 88, 91, 79
    ],

    "Marks": [
        85, 72, 90, 65, 78,
        88, 55, 76, 92, 70,
        84, 89, 80, 86, 73
    ],

    "Feedback": [
        "Excellent", "Good", "Excellent", "Average", "Good",
        "Excellent", "Poor", "Good", "Excellent", "Average",
        "Good", "Excellent", "Good", "Excellent", "Average"
    ]
}

df = pd.DataFrame(data)

# Display complete dataset
print("=" * 60)
print("DATASET")
print("=" * 60)
print(df)
print("\n" + "=" * 60)
print("BASIC INFORMATION")
print("=" * 60)

print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns.tolist())

print("\nData Types:")
print(df.dtypes)
```

```
print("\nFirst 5 Records:")
print(df.head())

print("\nLast 5 Records:")
print(df.tail())
print("\n" + "=" * 60)
print("DATA QUALITY CHECK")
print("=" * 60)

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Records:")
print(df.duplicated().sum())

print("\nUnique Values:")
print(df.nunique())
print("\n" + "=" * 60)
print("QUANTITATIVE ANALYSIS")
print("=" * 60)

# Select numerical columns
numerical_columns = [
    "Age",
    "Study_Hours",
    "Attendance",
    "Marks"
]

# Descriptive statistics
print("\nDescriptive Statistics:")
print(df[numerical_columns].describe())

# Mean
print("\nMean:")
print(df[numerical_columns].mean())

# Median
print("\nMedian:")
print(df[numerical_columns].median())

# Standard deviation
print("\nStandard Deviation:")
print(df[numerical_columns].std())
```

```
# Minimum and maximum
print("\nMinimum Values:")
print(df[numerical_columns].min())

print("\nMaximum Values:")
print(df[numerical_columns].max())
print("\n" + "=" * 60)
print("CORRELATION MATRIX")
print("=" * 60)

correlation = df[numerical_columns].corr()

print(correlation)
print("\n" + "=" * 60)
print("QUALITATIVE ANALYSIS")
print("=" * 60)

# Gender distribution
print("\nGender Distribution:")
print(df["Gender"].value_counts())

# Department distribution
print("\nDepartment Distribution:")
print(df["Department"].value_counts())

# Feedback distribution
print("\nFeedback Distribution:")
print(df["Feedback"].value_counts())

# Cross-tabulation
print("\nDepartment vs Gender:")
print(pd.crosstab(df["Department"], df["Gender"]))

print("\nDepartment vs Feedback:")
print(pd.crosstab(df["Department"], df["Feedback"]))
plt.figure(figsize=(8, 5))

sns.histplot(
    df["Marks"],
    bins=6,
    kde=True,
    color="royalblue"
)

plt.title("Distribution of Student Marks")
```

```
plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.tight_layout()
plt.show()
plt.figure(figsize=(8, 5))

sns.boxplot(
    y=df["Marks"],
    color="lightgreen"
)

plt.title("Box Plot of Student Marks")
plt.ylabel("Marks")
plt.tight_layout()
plt.show()
plt.figure(figsize=(8, 5))

sns.countplot(
    x="Department",
    data=df,
    hue="Department",
    palette="Set2",
    legend=False
)

plt.title("Number of Students by Department")
plt.xlabel("Department")
plt.ylabel("Number of Students")
plt.tight_layout()
plt.show()
```

OUTPUT:

```
==========================================================
DATASET
==========================================================
   Student_ID    Name  Age  Gender Department  Study_Hours  Attendance  \
0           1    Arun   20    Male        CSE            5         95
1           2    Bala   21    Male        ECE            3         80
2           3  Chitra   20  Female        CSE            6         98
3           4   Divya   22  Female        EEE            2         75
4           5   Eshan   21    Male        CSE            4         85
5           6   Farah   20  Female        ECE            6         92
6           7   Gokul   23    Male        EEE            1         60
7           8    Hari   21    Male        CSE            4         82
```

8	9	Isha	22	Female	ECE	7	96
9	10	Jaya	20	Female	EEE	3	78
10	11	Kiran	21	Male	CSE	5	90
11	12	Latha	22	Female	ECE	6	94
12	13	Manoj	20	Male	CSE	4	88
13	14	Nisha	21	Female	EEE	5	91
14	15	Omkar	22	Male	ECE	3	79

	Marks	Feedback
0	85	Excellent
1	72	Good
2	90	Excellent
3	65	Average
4	78	Good
5	88	Excellent
6	55	Poor
7	76	Good
8	92	Excellent
9	70	Average
10	84	Good
11	89	Excellent
12	80	Good
13	86	Excellent
14	73	Average

==========
BASIC INFORMATION
==========

Dataset Shape:
(15, 9)

Column Names:
['Student_ID', 'Name', 'Age', 'Gender', 'Department', 'Study_Hours', 'Attendance', 'Marks', 'Feedback']

Data Types:
Student_ID int64
Name object
Age int64
Gender object
Department object
Study_Hours int64
Attendance int64
Marks int64

```
Feedback       object
dtype: object

First 5 Records:
   Student_ID    Name  Age  Gender Department  Study_Hours  Attendance  Marks  \
0           1    Arun   20    Male        CSE            5          95     85
1           2    Bala   21    Male        ECE            3          80     72
2           3  Chitra   20  Female        CSE            6          98     90
3           4   Divya   22  Female        EEE            2          75     65
4           5   Eshan   21    Male        CSE            4          85     78

    Feedback
0  Excellent
1       Good
2  Excellent
3    Average
4       Good

Last 5 Records:
    Student_ID   Name  Age  Gender Department  Study_Hours  Attendance  Marks  \
10          11  Kiran   21    Male        CSE            5          90     84
11          12  Latha   22  Female        ECE            6          94     89
12          13  Manoj   20    Male        CSE            4          88     80
13          14  Nisha   21  Female        EEE            5          91     86
14          15  Omkar   22    Male        ECE            3          79     73

     Feedback
10       Good
11  Excellent
12       Good
13  Excellent
14    Average

==========================================================
DATA QUALITY CHECK
==========================================================

Missing Values:
Student_ID     0
Name           0
Age            0
Gender         0
Department     0
Study_Hours    0
Attendance     0
```

Marks 0
Feedback 0
dtype: int64

Duplicate Records:
0

Unique Values:
Student_ID 15
Name 15
Age 4
Gender 2
Department 3
Study_Hours 7
Attendance 15
Marks 15
Feedback 4
dtype: int64

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QUANTITATIVE ANALYSIS
==========

Descriptive Statistics:

	Age	Study_Hours	Attendance	Marks
count	15.000000	15.000000	15.000000	15.000000
mean	21.066667	4.266667	85.533333	78.866667
std	0.961150	1.667619	10.105633	10.453070
min	20.000000	1.000000	60.000000	55.000000
25%	20.000000	3.000000	79.500000	72.500000
50%	21.000000	4.000000	88.000000	80.000000
75%	22.000000	5.500000	93.000000	87.000000
max	23.000000	7.000000	98.000000	92.000000

Mean:
Age 21.066667
Study_Hours 4.266667
Attendance 85.533333
Marks 78.866667
dtype: float64

Median:
Age 21.0
Study_Hours 4.0
Attendance 88.0

Marks 80.0
dtype: float64

Standard Deviation:
Age 0.961150
Study_Hours 1.667619
Attendance 10.105633
Marks 10.453070
dtype: float64

Minimum Values:
Age 20
Study_Hours 1
Attendance 60
Marks 55
dtype: int64

Maximum Values:
Age 23
Study_Hours 7
Attendance 98
Marks 92
dtype: int64

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CORRELATION MATRIX
==========

	Age	Study_Hours	Attendance	Marks
Age	1.000000	-0.368396	-0.533402	-0.446948
Study_Hours	-0.368396	1.000000	0.944619	0.981516
Attendance	-0.533402	0.944619	1.000000	0.982540
Marks	-0.446948	0.981516	0.982540	1.000000

==========
QUALITATIVE ANALYSIS
==========

Gender Distribution:
Gender
Male 8
Female 7
Name: count, dtype: int64

Department Distribution:
Department

CSE 6
ECE 5
EEE 4
Name: count, dtype: int64

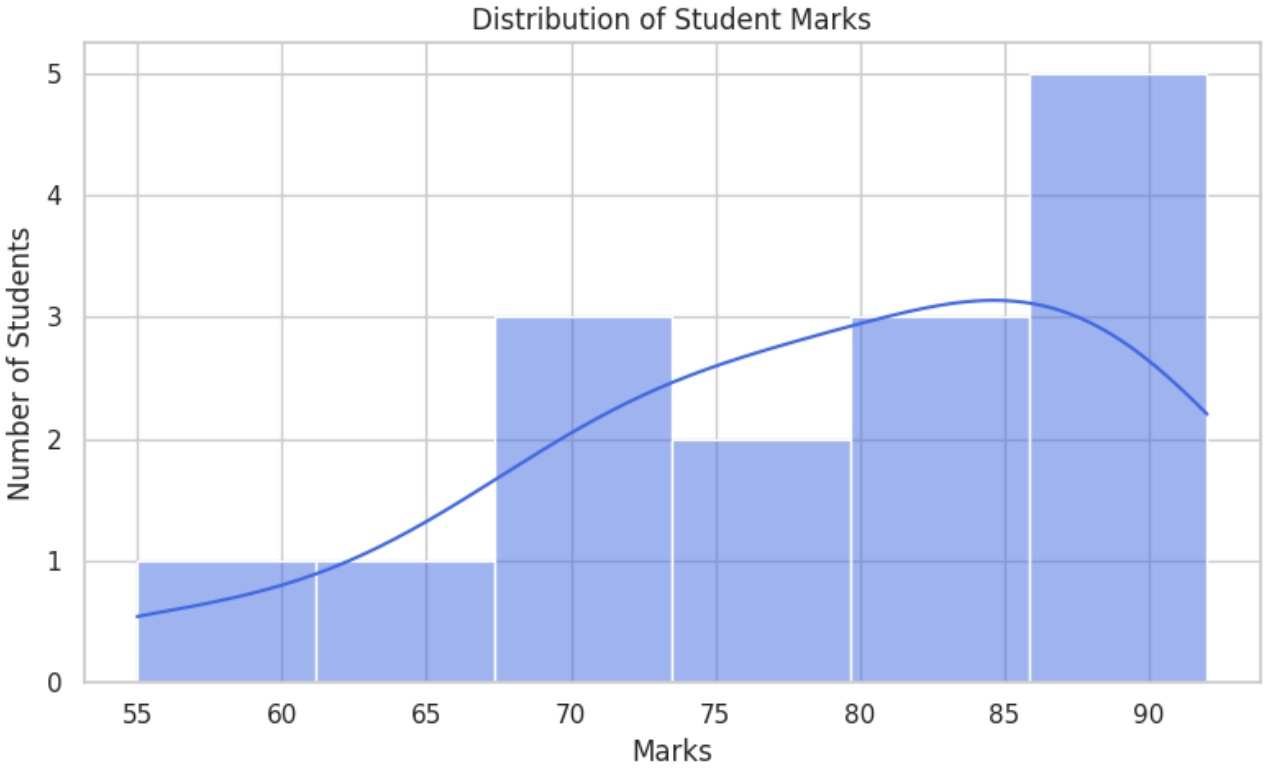
Feedback Distribution:
Feedback
Excellent 6
Good 5
Average 3
Poor 1
Name: count, dtype: int64

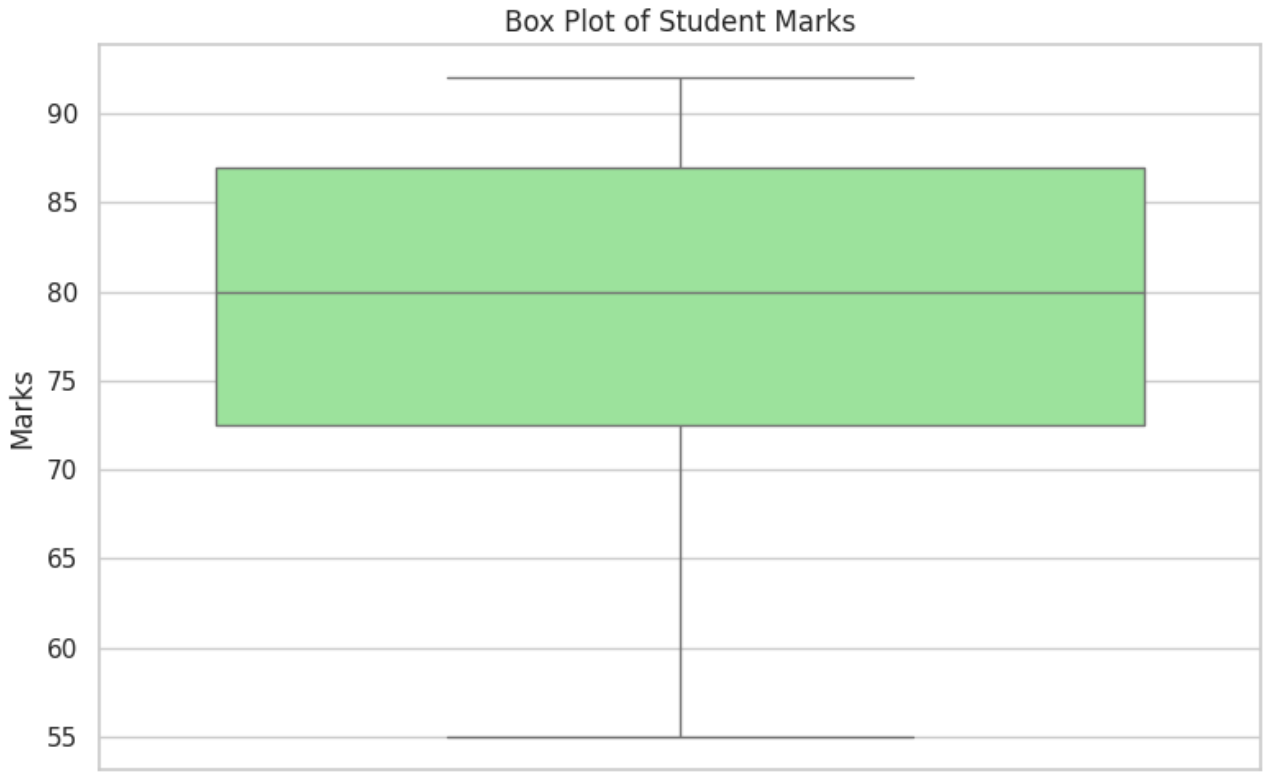
Department vs Gender:

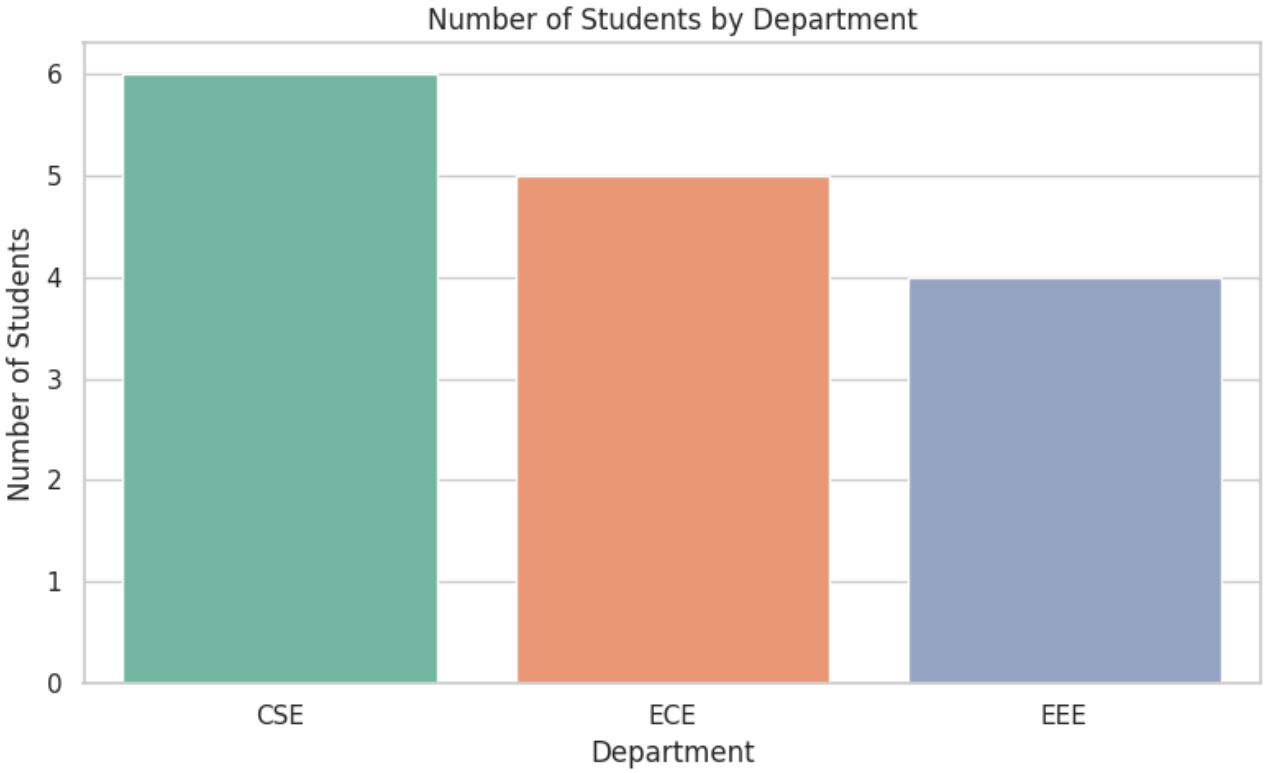
Department	Female	Male
CSE	1	5
ECE	3	2
EEE	3	1

Department vs Feedback:

Department	Average	Excellent	Good	Poor
CSE	0	2	4	0
ECE	1	3	1	0
EEE	2	1	0	1







RESULT:

The code is executed successfully.